SOFTWARE ARCHITECTURE

Client-Server architecture (cont.)

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Content

- Networking applications
- Remoting applications

Networking applications

- Transmission Control Protocol (TCP) application
- Web application
- File Transfer Protocol (FTP) application
- Mail application

TCP application

- What is TCP/IP?
  - TCP/IP is the communication protocol for communication between computers on the Internet.
  - TCP/IP stands for Transmission Control Protocol / Internet Protocol.
  - TCP/IP defines how electronic devices (like computers) should be connected to the Internet, and how data should be transmitted between them

TCP Uses a Fixed Connection

- TCP is for communication between applications.
- If one application wants to communicate with another via TCP, it sends a communication request. This request must be sent to an exact address. After a “handshake” between the two applications, TCP will set up a “**full-duplex**” communication between the two applications.
- The “full-duplex” communication will occupy the communication line between the two computers until it is closed by one of the two applications.
- User Datagram Protocol (UDP) is very similar to TCP, but simpler and less reliable.

IP is Connection-Less

- IP is for communication between computers.
- IP is a “connection-less” communication protocol.
- IP does not occupy the communication line between two computers.
  - IP reduces the need for network lines.
    - Each line can be used for communication between many different computers at the same time.
- With IP, messages (or other data) are broken up into small independent “packets” and sent between computers via the Internet.
- IP is responsible for “routing” each packet to the correct destination.

TCP/IP

- TCP/IP is TCP and IP working together.
- TCP takes care of the communication between your application software (i.e. your browser) and your network software.
- IP takes care of the communication with other computers.
- TCP is responsible for breaking data down into IP packets before they are sent, and for assembling the packets when they arrive.
- IP is responsible for sending the packets to the correct destination.
- TCP/IP uses 32 bits, or four numbers between 0 and 255, to address a computer.

Web application

- The Internet revolves around the client-server architecture.
- Your computer runs software called the **client** and it interacts with software known as the **server** located at a remote computer.
- The client is usually a browser such as Mozilla, Chrome, Firefox.
- Browsers interact with the server using a set of instructions called **protocols**.
- These protocols help in the accurate transfer of data through requests from a browser and responses from the server.

Internet protocols

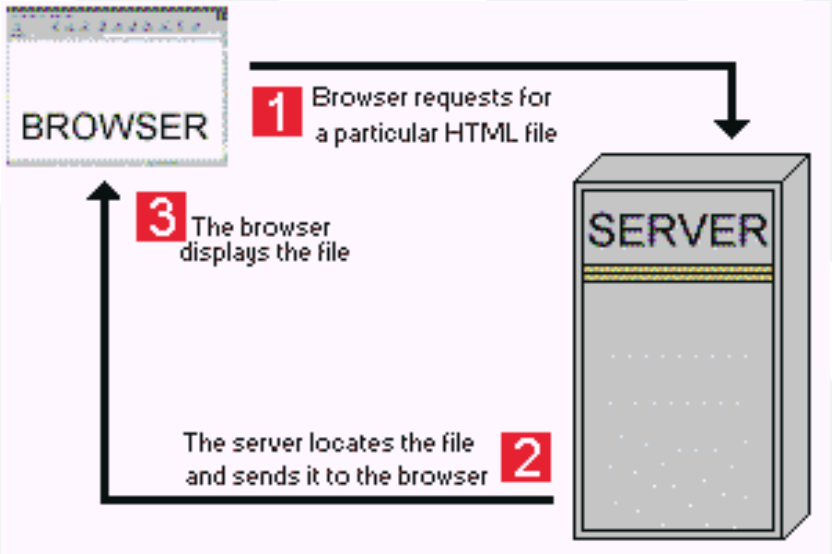
- There are many protocols available on the Internet.
  - HTTP (Hypertext transfer Protocol): used on the World Wide Web (WWW) for transferring web pages and files contained in web pages such as images.
  - FTP (File Transfer protocol): employed for transferring files from one machine to the other.
  - SMTP (Simple Mail Transport Protocol): used for email.
  - Telnet Protocol: Used to open telnet sessions.
- The web employs a connection-less protocol, which means that after every client-server interaction the connection between the two is lost

Client-server web models

- Static HTML pages
- Common Gateway Interface (CGI) Scripts
- Server-side scripting technologies

Static HTML pages

- The client (browser) requests for an HTML file stored on the remote machine through the server software.
- The server locates this file and passes it to the client.
- The client then displays this file on your machine.
- In this case, the HTML page is static.
- Static pages do not change until the developer modifies them.

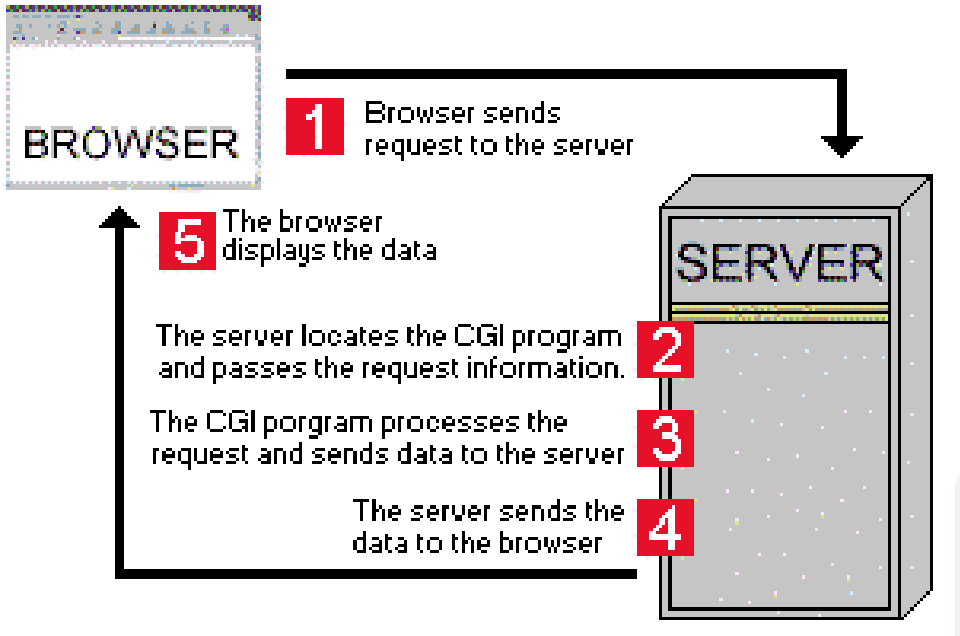


Common Gateway Interface (CGI) Scripts

- The Common Gateway Interface (CGI) provides the middleware between WWW servers and external databases and information sources.
- The World Wide Web Consortium (W3C) defined the Common Gateway Interface (CGI) and defined how a program interacts with a Hyper Text Transfer Protocol (HTTP) server.
- The Web server typically passes the form information to a small application program that processes the data and may send back a confirmation message.
- This process or convention for passing data back and forth between the server and the application is called the common gateway interface (CGI).

Common Gateway Interface (CGI) Scripts

- The server must do more work since CGI programs consume the server machine's processing power.
- The CGI program generates a dynamic HTML page. The contents of the dynamic page depend on the query passed to the CGI program.



Common Gateway Interface (CGI) Scripts

- Example
  - You type in the word ‘computers’ as the search query.
  - Your browser sends your request to the server.
  - The server checks the headers and locates the necessary CGI program and passes it the data from the request including your search query “computers”.
  - The CGI program processes this data and returns the results to the server.
  - The server then sends this formatted in HTML to your browser which in turn displays the HTML page.

Server-side scripting technologies

- Dynamic response generated using server-side technologies
- There are many server-side technologies today.
  - Active Server Pages (ASP): A Microsoft technology. ASP pages typically have the extension .asp.
  - Personal Home Pages (PHP): An open-source technology. PHP pages typically have .php, .phtml or .php3 file name extensions.
  - Java Server Pages: .jsp pages contain Java code.

Server-side scripting technologies

- These technologies make easier to maintain Web pages especially helpful for a large web site.
- The developer needs to embed the server-side language code inside the HTML page.
- This code is passed to the appropriate interpreter which processes these instructions and generates the final HTML displayed by the browser.
- Note, the embedded server-script code is not visible to the client (even if you check the source of the page) as the server sends ONLY the HTML code.

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Code inline vs Code behind

- Read these article
  - **ASP.NET Code-behind model overview** at https://learn.microsoft.com/en-us/troubleshoot/developer/webapps/aspnet/development/code-behind-model
  - **The Advantages of the Code-Behind Model in ASP.NET**, https://www.developer.com/microsoft/c-sharp/asp-net-code-behind-model/
- Share with us
  - What is the most difference between the two models?
  - In what situation, each model is more suitable?

File Transfer Protocol – FTP

- File Transfer Protocol (FTP) is a standard network protocol used to copy a file from one host to another over a TCP/IP-based network, such as the Internet.
- FTP is built on a client-server architecture and utilizes separate control and data connections between the client and server.
- FTP users may authenticate themselves using a clear-text sign-in protocol but can connect anonymously if the server is configured to allow it.
- The first FTP client applications were interactive command-line tools, implementing standard commands and syntax. Graphical user interface clients have since been developed for many of the popular desktop operating systems in use today.

The objectives of FTP

- Promote file (programs or data) sharing
- Efficiently transfer data from one computer to another
- Encourage indirect or implicit use of remote computers
- Provide a common platform for file storages among different hosts

FTP sites

- FTP sites are storehouses of hundreds to thousands of files – programs, data, music, video etc. In order to upload or download a file from an FTP site
- You need to know the login details – the FTP username and FTP password.
- Once connected, you can transfer files using the binary or ASCII mode.
- The binary mode is used for binary files like executable programs, music, images, video etc. while the ASCII mode is employed for plain text files such as those created in Notepad, vi etc.

What is anonymous FTP?

- On connection to an FTP site or FTP server, you would typically be asked the login details – a username and password.
- However, some severs enable what has come to be known as anonymous FTP.
- On Anonymous FTP servers you can log in with ‘anonymous’ as username and your email as password.
- You would then have access to all the data which has been placed in the anonymous FTP section.
- Example
  - ftp.gnu.org
  - ftp.pureftpd.org
  - ftp.vim.org
  - ftp.slackware.com

FTP and web sites

- The File Transfer Protocol is generally used by web developers to upload web sites to the host – the computer that stores the web site.
- For example, if you buy a web hosting package, the company will provide you with a username and password with which you'll connect to the FTP server to transfer your files.

File Transfer Protocol and the browser

- Modern browsers allow you to access sites using FTP.
- For example, https://www.infinityfree.net/

FTP clients

- An FTP Client is a dedicated program for FTP.
- There are many FTP clients available (and some of them are freeware).
- The interface of each program varies, but the information required is almost the same.
- You should know your username and password of the site (if it is not anonymous FTP) and the server address.
- Example: FileZilla at https://filezilla-project.org/

Email application

- Your email program uses different TCP/IP protocols:
  - It sends your emails using SMTP
  - It can download your emails from an email server using POP
  - It can connect to an email server using IMAP

Exercise

- Search over the Internet and find the addresses of these protocols of Gmail
  - STMP
  - POP
  - IMAP

SMTP – Simple Mail Transfer Protocol

- The SMTP protocol is used for the transmission of e-mails. SMTP takes care of sending your email to another computer.
- Normally your email is sent to an email server (SMTP server), and then to another server or servers, and finally to its destination.
- SMTP can only transmit pure text. It cannot transmit binary data like pictures, sounds or movies.
- SMTP uses the MIME protocol to send binary data across TCP/IP networks. The MIME protocol converts binary data to pure text.

POP – Post Office Protocol

- The POP protocol is used by email programs (like Microsoft Outlook) to retrieve emails from an email server.
- If your email program uses POP, all your emails are downloaded to your email program (also called email client), each time it connects to your email server.

IMAP – Internet Message Access Protocol

- The IMAP protocol is used by email programs (like Microsoft Outlook) just like the POP protocol.
- The main difference between the IMAP protocol and the POP protocol is that the IMAP protocol will not automatically download all your emails each time your email program connects to your email server.
- The IMAP protocol allows you to look through your email messages at the email server before you download them. With IMAP you can choose to download your messages or just delete them. This way IMAP is perfect if you need to connect to your email server from different locations, but only want to download your messages when you are back in your office.

Remoting applications

- What Is Remoting?
  - Remoting is the process of programs or components interacting across certain boundaries.
  - These contexts will normally resemble either different processes or machines.
  - In the .NET Framework, this technology provides the foundation for distributed applications—it simply replaces DCOM.

Remoting applications

- Allow to distribute business applications among various machines
- To improve performance, scalability, and maintainability.

Scenarios for .NET Remoting

- Centralized Business Logic
- Physical Separation of Layers
- Accessing Other Platforms

Centralized Business Logic

- One of the key scenarios for implementing remoting is the concentration of business logic on one or more central servers.
- This considerably simplifies the maintainability and operability of large-scale applications.
- Changes in business logic do not entail your having to roll out an application to your organization's 10,000 worldwide users—you just must update one single server.
- When this centralized business logic is shared among different applications, this labor-saving effect multiplies considerably; instead of patching several applications, you just must change the server's implementation.

Physical Separation of Layers

- The security of a company's vital databases represents a common concern in this time of Web-enabled businesses.
- The general recommendation is against directly connecting from the Web server to the database because this setup would allow attackers easy access to critical data after they have seized control of the Web server.
- Instead of this direct connection, an intermediate application server is introduced. This server is placed in a so-called demilitarized zone (DMZ), located between two firewalls.
- Because the application server doesn't allow the execution of arbitrary SQL statements yet provides object-oriented or function-based access to business logic, a security compromise of the Web server (which can only talk to the app server) is noncritical to a company's operations.

Accessing Other Platforms

- In today's mid- to large-scale enterprises, you will normally encounter a heterogeneous combination of different platforms, frameworks, and programming languages.
- It is not uncommon to find that a bunch of tools have been implemented: Active Server Pages (ASP), Java Server Pages (JSP), PHP, or ColdFusion for Web applications, Visual Basic or Java for in-house applications, C++ for server-side batch jobs, scripting languages for customizing CRM systems, and so on.
- Integrating these systems can be a daunting task for system architects.
- Remoting architectures like CORBA, SOAP, and .NET Remoting are an absolute necessity in large-scale enterprise application integration.

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Q&A

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